Solutions for Exercises
Rigid Analytic Geometry

So Murata

SoSe 26, University of Bonn

Sheet1. Banach Spaces over an Ultrametric Field

Definition 1.1. *Let K be a field together with an absolute value $|\cdot|_K$ such that for each $x, y \in K$,*

i. $|x+y|_K \leq \max\{|x|_K, |y|_K\}$

ii. $|xy|_K = |x|_K|y|_K$

iii. $|0|_K = 0, |1|_K = 1$

iv. (K, d) *where* $d(x,y) = |x-y|_K$ *is a complete non-discrete metric space.*

We further denote,

$$K^\circ = \{x \in K \mid |x|_K \leq 1\},$$
$$K^{\circ\circ} = \{x \in K \mid |x|_K < 1\}.$$

Problem 1.1. *Show that* $\mathrm{Spec}(K^\circ) = \{(0), K^{\circ\circ}\}$.

Solution. Let $x \in K$ be such that $x \notin K^\circ$, then $|x|_K > 1$. Then we have,

$$|x^{-1}|_K = 1/|x|_K < 1.$$

Therefore $x^{-1} \in K^\circ$. Thus K° is a valuation ring of K. Therefore $\mathrm{Spec}(K^\circ)$ consists of two points. □

Definition 1.2. *Let K be as above. A K-Banach space E is a ultrametric complete K-vector space. A subset X of E is bounded if $\|X|E\|$ is bounded in $\mathbb{R}$.*

Definition 1.3. *A K-Banach algebra is a K-algebra A equipped with K-Banach structure together with*

$$\|ab|A\| \leq \|a|A\|\|b|A\|, \|1|A\| \leq 1$$

It is called multiplicative if $\|ab|A\| = \|a|A\|\|b|A\|$. It is power-multiplicative if $\|a^n|A\| = \|a|A\|^n$.

Definition 1.4. *A Tate algebra with n-variables is*

$$\mathbb{T}_n = \left\{ f = \sum_{\alpha \in \mathbb{Z}^n_{\geq 0}} f_\alpha X^\alpha \in K[\![X_1, \cdots, X_n]\!] \middle| \forall \varepsilon >_{\mathbb{R}} 0, |\{f_\alpha | |f_\alpha|_K > \varepsilon\}| < \infty \right\}.$$

Let us define the norm on $\mathbb{T}_n$ by

$$\|f|\mathbb{T}_n\| = \max_{\alpha \in \mathbb{Z}_{\geq 0}} |f_\alpha|_K.$$

Problem 1.2. *Show that $\mathbb{T}_n$ is multiplicative.*

Solution. Let $f, g \in \mathbb{T}_n$ and $\alpha_0, \beta_0 \in \mathbb{Z}^n_{0\geq}$ such that

$$\|f|\mathbb{T}_n\| = |f_{\alpha_0}|_K, \|g|\mathbb{T}_n\| = |g_{\beta_0}|_K,$$

and they are minimum among these indices. By using the ultrametricity,

$$\left|\sum_{\gamma=\alpha+\beta} f_\alpha g_\beta\right| \leq \max_{\gamma=\alpha+\beta} |f_\alpha g_\beta|_K \leq |f_{\alpha_0} g_{\beta_0}|.$$

By the minimality, we have,

$$\forall(\alpha,\beta), \alpha+\beta = \alpha_0+\beta_0 \Rightarrow |f_\alpha g_\beta|_K \leq |f_{\alpha_0}||g_{\beta_0}|_K.$$

Observe that

$$\begin{aligned}
|f_{\alpha_0} g_{\beta_0}| &= \left|(fg)_{\alpha_0+\beta_0} - \sum_{\alpha+\beta=\alpha_0+\beta_0} f_\alpha g_\beta\right|, \\
&\leq \max\left\{|(fg)_{\alpha_0+\beta_0}|, \left|\sum_{\alpha+\beta=\alpha_0+\beta_0} f_\alpha g_\beta\right|\right\}, \\
&|f_{\alpha_0} g_{\beta_0}|. \\
\Rightarrow &|f_{\alpha_0} g_{\beta_0}| = |(fg)_{\alpha_0+\beta_0}|.
\end{aligned}$$

□

Definition 1.5. *Let A be a K-Banach algebra. Define,*

$$\begin{aligned}
A^\circ &= \{a \in A \mid \{a^n | n \in \mathbb{N}\} \text{ is bounded in } A\}, \\
A^{\circ\circ} &= \{a \in A \mid \lim_{k\to\infty} a^k = 0\}.
\end{aligned}$$

Problem 1.3. *Show that A° is a subring and $A^{\circ\circ}$ is an ideal satisfying $\sqrt{A^{\circ\circ}} = A^{\circ\circ}$.*

Solution. Let $a, b \in A^\circ$ and M_a, M_b be such that

$$\{\|a^n|A\| | n \in \mathbb{N}\} \subseteq [0, M_a], \{\|b^n|A\| | n \in \mathbb{N}\} \subseteq [0, M_b].$$

Then,

$$\|ab|A\| \leq \|a|A\|\|b|A\| \Rightarrow \|(ab)^n|A\| \leq \|a|A\|^n \|b|A\|^n \leq M_a M_b.$$

Similarly, by the ultrametricity, we have A° is closed under addition and multiplication. As it contains $1, 0$, it is a subring.

For the latter part, let $x \in A^{\circ\circ}$, then,

$$\|(ax)^k|A\| \overset{k\to\infty}{\to} 0.$$

We then know that the sequence $((ax)^k)_{k\in\mathbb{N}}$ is a Cauchy sequence thus has a limit which has norm 0 thus converges to 0. To show that this is closed under addition, note that $\{(x+y)^k\}_{k\in\mathbb{N}}$ is Cauchy and the limit of the norm is 0.

Finally, $(a^k)^l \overset{l\to\infty}{\to} 0$ then again using the completeness we finish the solution. □

Problem 1.4. *If A is power-multiplicative then,*

$$\begin{aligned}
A^\circ &= \{a \in A \mid \|a|A\| \leq 1\}, \\
A^{\circ\circ} &= \{a \in A \mid \|a|A\| < 1\}.
\end{aligned}$$

Solution. The definitions of these sets are now,

$$\begin{aligned}
A^\circ &= \{a \in A \mid \{\|a|A\|^n | n \in \mathbb{N}\} \text{ is bounded.}\}, \\
A^{\circ\circ} &= \{a \in A \mid \|a|A\|^n \overset{n\to\infty}{\to} 0\}.
\end{aligned}$$

And the suggested forms are the paraphrases of these. □

Problem 1.5. *A K-Banach algebra A admits no proper open ideal.*

Solution. Let $I \subseteq A$ be an open ideal. Then I is a neighborhood of 0. Let $a \in K^{\circ\circ}$, then $a \cdot 1_A \in A^{\circ\circ}$, thus for large enough k, $(a \cdot 1_A)^k \in I$. But $a^{-1}1_A$ is an inverse to it thus I is not proper. □

Problem 1.6. *The group of unit of a K-Banach algebra A is a neighborhood of* 1.

Solution. Let $U = \|^{\|} - 1([0,1))$, then this is open. Note that $1 + U$ is contained in the group of units as every element of U is topologically nilpotent. By the opennes of addition map, we conclude $1 + U$ is a open neighborhood of 1 contained in such group. □

Problem 1.7. *Show K-Banach algebra has no proper dense ideal.*

Proof. Suppose I is dense, then I intersects with the group of units thus I is the whole. □

Sheet2. Basics of Spectral Spaces

Definition 2.1. *Let V be a K-Banach space and $\|\cdot\|_1, \|\cdot\|_2$ be norms. These two are said to be equivalent if there exists a constant $C \geq 1$ such that*

$$\forall v \in V, \|v\|_1 \leq C\|v\|_2 \leq C^2\|v\|_2.$$

Definition 2.2. *The n-dimensional K-vector space K^n will have the structure of a K-Banach space under the following norm,*

$$\|(x_i)_{i=1}^n | K^n\| = \max_{1 \leq i \leq n} |x_i|_K.$$

Problem 2.1. *Show that for any K-norm $\|\cdot\|_1$ on K^n, there is $C \geq 1$ such that*

$$\forall x \in K^n, \|x\|_1 \leq C\|x\|.$$

Solution. Let $e_1, \cdots, e_n \in K^n$ be the standard basis elements. Then for any $x = \sum_{i=1}^n x_i e_i$ ¡

$$\|x\|_1 \leq \max_{1 \leq i \leq n} |x_i|_K \|_{\|} 1 e_i \leq \max_{1 \leq i \leq n} \|e_i\| \|x\|.$$

Thus take $\max_{1 \leq i \leq n} \|e_i\| = C$. □

Problem 2.2. *Show that there is $C \geq 0$ such that for any $x \in K^n$,*

$$\|x\| \leq \|x\|_1.$$

Solution. Suppose such C does not exists. Then we can take a sequence $(x_n) \in K^n$ such that

$$\|x_n\| > n\|x_n\|_1.$$

Now set $y_n = \frac{x_n}{\|x_n\|}$. Then we have,

$$\|y_n\|_1 = \frac{\|x_n\|_1}{\|x_n\|} < \frac{1}{n} \overset{n \to \infty}{\to} 0.$$

This shows a contradiction as $\|y_n\| = 1$, that is, there is an entry whose absolute value is 1. □

Problem 2.3. *Show that the previous inequality fails if K is not complete with respect to the induced metric from $|\cdot|_K$.*

Solution. Let $\widehat{K}$ be a completion of K with respect to the metric and take $\alpha \in \widehat{K} \backslash K$. Define a norm on K^2 by,

$$\|(a,b)\|_1 = |a + b\alpha|_{\widehat{K}}.$$

This indeed defines a norm. Let $(x_n)_{n \in \mathbb{N}}$ be a seuqence of K converging to $-\alpha$. Furthermore, take,

$$v_n = (x_n, 1).$$

Then $\|v_n\| \geq 1$. But $\|v_n\|_1 \to 0$ as $n \to \infty$. □

Problem 2.4. *Let L/K be a finite field extension then there is $C > 0$ such that*

$$\forall x, y \in L, \|xy\| \leq C\|x\|\|y\|.$$

Solution. Let $m_x \in \mathrm{Hom}(L, L)$ be such that

$$m_x(y) = xy.$$

This is a linear operator in a finite dimensional vectorspace. Therefore the operator norm is bounded. Furthermore the assignment,

$$L \ni x \mapsto m_x \in \mathrm{Hom}(L, L),$$

is also a linear operator of finite dimensional vectorspaces. Thus there is $C > 0$ such that

$$\|m_x\|_{\mathrm{Op}} \leq C\|x\|.$$

Therefore, we obtain,

$$\|xy\| \leq \|m_x\|_{\mathrm{Op}}\|y\| \leq C\|x\|\|y\|.$$

□

Problem 2.5. *Let A be a ring and $\mathcal{U}$ be an ultrafilter on $\mathrm{Spec}(A)$. Show that*

$$\mathfrak{g}_{\mathcal{U}} = \{x \in A \mid V(x) \in \mathcal{U}\},$$

for any $x \in \mathfrak{g}_{\mathcal{U}}$ and $a \in A$, $ax \in \mathfrak{g}_{\mathcal{U}}$.

Solution. $V(ax) = V(a) \cup V(x) \supseteq V(x)$, thus $V(ax) \in \mathcal{U}$. □

Definition 2.3. *Let A be a ring. A cone C of A is a subset such that*

i. $A^2 \subseteq C$,

ii. $x, y \in C \Rightarrow x + y \in C, xy \in C$.

C is called proper if $-1 \notin C$. It is prime if

$$\forall a, b \in A, ab \in C, a \notin C \Rightarrow -b \in C.$$

Definition 2.4. *We denote the set of all prime cones of A to be $\mathrm{Sper}(A)$. For $a \in A$, we denote,*

$$\mathcal{P}(a) = \{P \in \mathrm{Sper}(A) \mid -a \notin P\}.$$

Furthermore, we set,

$$\mathcal{P}(a_1, \cdots, a_n) = \bigcap_{i=1}^{n} \mathcal{P}(a_i).$$

We conventionally set $\mathcal{P}(\emptyset) = \mathrm{Sper}(A)$.
We further denote,

$$\mathcal{B}_{\mathrm{Sper}(A)} = \{\mathcal{P}(a_1, \cdots, a_n) \mid a_1, \cdots, a_n \in A\}.$$

Lemma 2.1. *Let $\mathcal{B}$ be a topology base of the topological space X.*

a) Let $\mathcal{U}$ be an ultrafilter on X and $g \in X$ such that for all $\Omega \in \mathcal{B}$, $g \in \Omega \Leftrightarrow \Omega \in \mathcal{U}$. Then g is a generic limit of $\mathcal{U}$.

b) If X is T_0 and for every ultrafilter $\mathcal{U}$ on X, there is $g \in X$ as in a, then X is spectral and the elements of $\mathcal{B}$ are quasi-compact.

Problem 2.6. *Show that $\mathrm{Sper}(A)$ is spectral and elements of $\mathcal{B}_{\mathrm{Sper}(A)}$ are quasi-compact.*

Solution. Denote $X = \mathrm{Sper}(A)$. For $x, y \in X$, if $x \neq y$, then there is $a \in A$ such that $a \in x$ but $a \notin y$. Therefore, $\mathcal{P}(-a)$ will sepaarte them.

In order to use Lemma 2.1, we show that for an ultrafilter $\mathcal{U}$ on X,

$$\mathfrak{g}_{\mathcal{U}} = \{a \in A \mid \{P \in X \mid x \in P\} \in \mathcal{U}\}$$

is the g in Lemma 2.1. First we need to show that this is a prime cone.For $a \in A$, let us denote,

$$F(a) = \{P \in X \mid a \in P\}.$$

With this, we can rewrite,

$$\mathfrak{g}_{\mathcal{U}} = \{a \in A \mid F(a) \in \mathcal{U}\}.$$

Since $\mathcal{U}$ is a filter and

$$F(-1) = \emptyset.$$

Thus $-1 \notin \mathfrak{g}_{\mathcal{U}}$. If $x, y \in \mathfrak{g}_{\mathcal{U}}$, then

$$F(x+y) \supseteq F(x) \cap F(y), F(xy) \supseteq F(x) \cap F(y).$$

Thus by the filter property, this is closed under addition and multiplication. Finally, if $xy \in \mathfrak{g}_{\mathcal{U}}$ and $x \notin \mathfrak{g}_{\mathcal{U}}$, then

$$F(-y) \supset F(xy) \cap (X \backslash F(x)).$$

Using the ultrafilter property and $F(x) \notin \mathcal{U}$, we have $X \backslash F(x) \in \mathcal{U}$. Thus $-y \in \mathfrak{g}_{\mathcal{U}}$.

To show that $\mathfrak{g}_{\mathcal{U}}$ is the generic limit, first observe that

$$F(-a) \cup \mathcal{P}(a) = X.$$

Therefore,

$$\begin{aligned} \mathfrak{g}_{\mathcal{U}} \in \Omega &\iff \mathfrak{g}_{\mathcal{U}} \in \bigcap_{i=1}^{n} (X \backslash F(-a_i)), \\ &\iff F(-a_1), \cdots, F(a_n) \notin \mathcal{U}, \\ &\iff \mathcal{P}(a_1), \cdots, \mathcal{P}(a_n) \in \mathcal{U}. \end{aligned}$$

The last equivalence is due to the ultrafilterness of $\mathcal{U}$. By Lemma 2.1, X is spectral and $\mathcal{B}_X$ consists of quasi-compact elements. □

Problem 2.7. *Let X be a set and $\eta \in X$. We give a structure of a topological space to X such that open sets U are either emptyset or $\eta \in U$ and $X \backslash U$ is finite. Show that X is spectral.*

Solution. Let $x, y \in X$ be distinct points such that $y \neq \eta$. Then $X \backslash y$ will separate x and y thus it is T_0.

Observe that every closed subsets are finite set of points. Therefore, every irreducible closed subset is a single point. Thus the sobriety is now shown.

Let U be an open set and $(U_n)_{n \in \mathbb{N}}$ be an open cover. Then, $\bigcup_{n \in \mathbb{N}} U_n$ is again open therefore exists a finite set F such that

$$U \subseteq \bigcup_{n=1}^{\infty} U_n = X \backslash F.$$

Clearly, we can pick finite subcollection such that each element does not contain F. Thus U is quasi-compact. □

Problem 2.8. *Inherit X from the problem above. Describe X_{con} by giving neighborhood basis for each $x \in X$.*

Solution. Suppose $x \neq \eta$. Since $X \backslash \{x\}$ is open, $\{x\}$ is open in X_{con}. For η, if $X \backslash U \ni \eta$, then $U = \emptyset$. Therefore the neighborhood basis of η is the set of all non-empty open sets. □

Sheet3. Introduction to Valuations

Problem 3.1. *Let L/K be a finite field extension. Then show that there is a K-norm $\|\cdot|L\|$ on L such that*

$$\forall x, y \in L, \|xy|L\| \leq \|x|L\|\|y|L\|.$$

Solution. By Problem 2.4, there is a constant $C > 0$ for every norm $\|\cdot\|$ on L such that

$$\|xy\| \leq C\|x\|\|y\|.$$

Set $\|\cdot|L\| = C\|\cdot\|$. Then,

$$\|xy|L\| = C\|xy\| \leq C^2\|x\|\|y\| = \|x|L\|\|y|L\|.$$

□

Definition 3.1. *A sequence $(a_n)_{n\in\mathbb{N}} \subseteq \mathbb{R}$ is said to be subadditive if*

$$\forall m, n \in \mathbb{N}, a_{m+n} \leq a_m + a_n.$$

Theorem 3.1 (Fekete's Lemma)**.** *Let $(a_n)_{n\in\mathbb{N}}$ be a subadditive sequence. Then we have,*

$$\lim_{n\to\infty} \frac{a_n}{n} = \inf_{n\in\mathbb{N}} \frac{a_n}{n},$$

possibly the infimum being $-\infty$.

Problem 3.2. *For every norm $\|\cdot|L\|$ on the K-vector space L, show that*

$$|l|_L = \lim_{n\to\infty} \sqrt[n]{\|l|L\|}$$

is well-defined and independent of the choice of the norm.

Solution. We have seen that every norm is equivalent to $\|\cdot\|_\infty$ on L. Thus independence follows from that

$$c\|l\|_\infty \leq \|l|L\| \leq C\|l\|_\infty,$$

then,

$$\sqrt[n]{c}\|l^n\|_\infty \leq \|l^n|L\| \leq \sqrt[n]{C}\|l^n\|_\infty.$$

By the sandwich theorem, the independence follows. For the well-definedness, by the property of norms, recall we have,

$$\|ab\|_\infty \leq \|a\|_\infty\|b\|_\infty.$$

Thus set $a_n = \log\|l^n\|_\infty$. Then this is subadditive and $\frac{a_n}{n}$ converges as $n \to \infty$. It remains to show that for any $k \in K, l \in L$, $|kl|_L = |k|_K|l|_L$. □

Problem 3.3. *Show that for any $a, b \in L$, $|ab|_L \leq |a|_L|b|_L$ holds.*

Solution. Follows immediately from the property of limit. □

Problem 3.4. *Show that for such $|\cdot|_L$, we have for any $k \in K \subseteq L$,*

$$|k|_L = |k|_K.$$

Solution.

$$|k|_L = \lim_{n\to\infty} \sqrt[n]{\|k^n\|_\infty} = \lim_{n\to\infty} \sqrt[n]{|k|_K^n} = |k|_K.$$

□

Problem 3.5.

Problem 3.6.

Definition 3.2. *Let X be a spectral space and $x \in X$. We set,*

$$S_x = \{y \in X \mid y \rightsquigarrow x\}.$$

Problem 3.7. *Show that for $x, y \in X$ for a spectral space X, we have,*

$$S_x \cap S_y = \bigcap_{\substack{\Omega \in \mathfrak{Q}_C(X) \\ x \in \Omega}} \bigcap_{\substack{\Theta \in \mathfrak{Q}_C(X) \\ y \in \Theta}} \Omega \cap \Theta.$$

Solution. Since X is spectral $\mathfrak{Q}_C(X)$ is a topology base of X. Therefore, for $z \in X$,

$$z \rightsquigarrow x \iff [\forall \Omega \in \mathfrak{Q}_C(X), z \in \Omega \Rightarrow x].$$

Therefore, we obtain,

$$S_x = \bigcap_{x \in \Omega} \Omega.$$

□

Corollary 3.1. *For a spectral space X, the following statements are equivalent.*

1. *X is T_0.*
2. *$X = X_{\mathrm{con}}$ as topological space.*
3. *Every quasi-compact subset of X is closed.*
4. *X is Hausdorff.*
5. *X is T_4.*

Problem 3.8. *Show that for a spectral space X, $x, y \in X$ admit disjoint neighborhoods if and only if $S_x \cap S_y = \emptyset$.*

Solution. $\Rightarrow$ is immediate from the previous problem. For the other direction, observe that X_{con} is compact and

$$X = \bigcup_{\substack{x \in \Omega \\ y \in \Theta}} (X \backslash (\Omega \cap \Theta)).$$

By Corollary 3.1, X_{con} is Hausdorff, in particular, it is compact. Therefore, we obtain,

$$X = \bigcap_{i=1}^{n} (X \backslash (\Omega_i \cap \Theta_i)).$$

As $\mathfrak{Q}_C(X)$ is closed under finite intersection, we have shown the statement. □

Problem 3.9. *Let $v : A \to \Gamma \cup \{0\}$ be a valuation and H be a convex subgroup of Γ containing all $v(a) \geq 1$ for $a \in A$. Show that*

$$w(a) = \begin{cases} v(a), & (w(a) \in H), \\ 0, & (\ \textit{otherwise}\ ) \end{cases}$$

is a valuation of A.

Solution. Clearly $w(1) = 1, w(0) = 0$. Let $a, b \in A$ then there are four cases that $v(a), v(b)$ belong to H or not. If at least one of them belong to H, then we have the desired form by the cancellation law. Suppose neither belong to Γ. If $v(ab) \in H$, then there is $c \in A$ such that $v(c^{-1}) \leq v(ab) \leq v(c)$. □

Sheet4. Tate Algebra

Proposition 4.1. *Let L/K be a finite extension of fields and $\mathbb{T}_m$ be the Tate algebra in m variables $x_1, \cdots, x_m$. For any $\mathfrak{m} \subseteq \mathbb{T}_m$ a maximal ideal there is $(x_1, \cdots, x_m) \in (L^\circ)^m$ such that*

$$\mathfrak{m} = \mathfrak{m}_x := \{f(x_1, \cdots, x_m) \mid f \in \mathbb{T}_m\}.$$

Problem 4.1. *Recall that there is a unique absolute value on L extended from K. Show that for any $f \in \mathbb{T}_m$,*

$$\forall x \in (L^\circ)^m, |f(x)| \leq \|f|\mathbb{T}_m\|.$$

Solution. Follows immediately from the fact that $|x_i| \leq 1$. □

Problem 4.2. *Let $f \in \mathbb{T}_m$. Show that the following are equivalent.*

(i). For any finite extension L/K and $x \in (L^\circ)^m$, $f(x) \neq 0$.

(ii). $f \in \mathbb{T}_m^\times$

(iii). For any finite extension L/K and $x \in (L^\circ)^m$, $f(x) \neq 0$ and $|f(x)| = \|f|\mathbb{T}_m\|$.

Solution. (i)⇒(ii). Suppose $f \notin \mathbb{T}_m^\times$, then there is a maximal ideal $\mathfrak{m}$ containing f. By Proposition 4.1, there is a finite extension L/K and $x \in (L^\circ)^m$ such that $f(x) = 0$.

(ii)⇒(iii). The first part is obvious. Remains to show the latter. Let $g = f^{-1} \in \mathbb{T}_m^\times$. Then

$$\|f|\mathbb{T}_m\|^{-1} \leq |g(x)| \leq \|g|\mathbb{T}_m\| = \|f|\mathbb{T}_m\|^{-1}.$$

(iii)⇒(i) is trivial. □

Problem 4.3. *Let K be algebraically closed, $x \in K^\circ$ and $f \in \mathbb{T}_1$. Suppose there is $R \in (0, 1]$ such that f takes no zeros on,*

$$B = K_{\leq R}(x) = \{y \in K \mid |x - y| < R\}.$$

Show that $y \mapsto |f(y)|$ is constant on B.

Solution. By setting $f(x) \mapsto f\left(\frac{x}{R}\right)$, we may assume that $R = 1$. Also by shifting, we also assume that $x = 0$. Then by the previous problem $|f(y)| = \|f|\mathbb{T}_1\|$. □

Problem 4.4. *Let L/K be an algebraic extension which is perfect. Then the completion $\mathbb{L}$ of L with respect to the absolute value $|\cdot|_L$ is perfect.*

Solution. Suppose K is of characteristic p. Let $x \in \mathbb{L}$. Then there is a sequence $(x_i)_{i\in\mathbb{N}}$ in L converging to x. Since L is perfect, there is $(y_i)_{i\in\mathbb{N}}$ such that $y_i^p = x_i$. Then

$$x_i - x_j = y_i^p - y_j^p = (y_i - y_j)^p.$$

Therefore

$$|y_i - y_j| = |x_i - x_j|^{\frac{1}{p}}.$$

Thus (x_i) is Cauchy then so is (y_i). Thus this has a limit y in $\mathbb{L}$ satisfying $y^p = x$. □

Remark 4.1. *By the uniqueness of the induced absolute value $|\cdot|$ on L from $|\cdot|_K$ in K, if $\sigma : L \to L$ is an automorphism, we have $|\cdot| = |\sigma(\cdot)|$.*

Problem 4.5. *Let L/K be Galois and $\alpha, \beta \in L$. Suppose for any $\tilde{\alpha}$ which is a conjugate of α not equal to it,*

$$|\beta - \alpha| < |\beta - \tilde{\alpha}|.$$

Show that $\alpha \in K(\beta)$.

Solution. We first show that in this setting $\sigma \in \mathrm{Gal}(L/K)$ fixes β then σ fixes α too.

$$|\beta - \sigma(\alpha)| = |\sigma(\beta) - \sigma(\alpha)| = |\beta - \alpha|.$$

Therefore $\sigma(\alpha) = \alpha$. By Galois theory, we have,

$$K(\beta) = L^{\mathrm{Gal}(L/K(\beta))}.$$

Therefore $\alpha \in K(\beta)$. □

Problem 4.6. *Let $(\alpha_i)_{1\leq i\leq d} \in K^d$ and define $(p_i)_{0\leq i\leq d}$ to be such that*

$$f(T) = \prod_{i=1}^{d}(T - \alpha_i) = \sum_{i=0}^{d} p_i T^i.$$

Show that

$$\max\{|\alpha_i|\}_{i=1,\cdots,d} = \max\{\sqrt[d-i]{|p_i|_K} \mid i = 0, \cdots, d-1\}.$$

Solution. Let $R = \max\{|\alpha_i|\}_{i=1,\cdots,d}$. Since we have,

$$p_i = (-1)^{d-i}\left(\sum_{|I|=d-i}\prod_{i\in I}\alpha^i\right).$$

Therefore by the ultrametricity, we have,

$$\max\{\sqrt[d-i]{|p_i|_K} \mid i = 0, \cdots, d-1\} \leq R.$$

On the other hand, suppose that the strict inequality holds in the above. Then there is k such that

$$\max\{\sqrt[d-i]{|p_i|_K} \mid i = 0, \cdots, d-1\} < |\alpha_k|_K.$$

Let $S = \max\{\sqrt[d-i]{|p_i|_K} \mid i = 0, \cdots, d-1\}$, then,

$$|f(\alpha_k)|_K = \left|\alpha_k^d + \sum_{i=0}^{d-1} S^{d-i}\alpha_K^i\right| = |\alpha_k|^d \neq 0.$$

Thus a contradiction. □

Sheet5. Grothendieck Topology

Problem 5.1. *Let $R \subseteq K$ be a valuation ring. Then show that for $x, y \in K$, either $x \in yR$ or $y \in xR$.*

Solution. Suppose $x \notin yR$, then $\frac{x}{y} \notin R$, thus $\frac{y}{x} \in R$ thus $y \in xR$. □

Notation 5.1. *Let*

$$\mathfrak{m}_R = \{0\} \cup \{x \in K^\times \mid x^{-1} \notin R\}.$$

Problem 5.2. *Show that $(R, \mathfrak{m}_R)$ is a local ring.*

Solution. Suppose $x, y \in \mathfrak{m}_R$, then $x^{-1}(x + y) \in \mathfrak{m}_R$ $R\backslash R^\times = \mathfrak{m}_R$ is clear by definition. □

Problem 5.3. *A ring is local if and only if $R\backslash R^\times$ is an ideal.*

Solution. □

Problem 5.4. *Let $R \subseteq K$ be a valuation ring then R is integrally closed.*

Solution. Suppose $x \in K$ is integral over R, then,

$$x^n + a_{n-1}x^{n-1} + \cdots + a_0 = 0$$

for some $a_{n-1}, \cdots, a_0 \in R$. If $x \notin R$, then $x^{-1} \in R$. Therefore,

$$x = -(a_{n-1} + a_{n-2}x^{-1} + \cdots + a_0 x^{n-1}) \in R.$$

This is a contradiction. □

Problem 5.5. *Ideals of a valuation ring R are linearly ordered by inclusion.*

Solution. Let $I, J \subsetneq R$ be ideals such that $I \subsetneq J$. Let $x \in I \backslash J$. Then for any $y \in J$, either $x \in yR$ or $y \in xR$. But $x \notin J$ means $y \in xR$. Therefore $J \subseteq I$. □

Problem 5.6. *Every finitely generated ideal in a valuation ring is principal.*

Solution. Trivial from the previous problem. □

Problem 5.7. *An ideal of a valuation ring is prime if and only if it is non-trivial and radical.*

Solution. One direction is clear. Suppose $\mathfrak{p} \subseteq R$ is non-trivial and radical. Suppose $xy \in \mathfrak{p}$ and assume $y \in xR$ or equivalently $y = rx, r \in R$. Then $r \cdot xy \in \mathfrak{p}$ thus $y^2 \in \mathfrak{p}$ and using that it is radical $y \in \mathfrak{p}$. □

Proposition 5.1. *Let A be a ring and $A \xrightarrow{v} \Gamma \cup \{0\}$ be a valuation. Then*

$$A/\mathfrak{p} \ni a + \mathfrak{p} \mapsto v(a) \in \Gamma \cup \{0\}$$

defines a valuation. Furthermore,

$$\mathrm{Frac}(A/\mathfrak{p}) \ni \frac{a}{b} \mapsto \frac{v(a)}{v(b)} \in \Gamma \cup \{0\}$$

also defines a valution.

Definition 5.1. *Let A be a ring we denote,*

$$V_A = \{(\mathfrak{p}, R) \mid \mathfrak{p} \in \mathrm{Spec}(A),\ R \text{ is a valuation ring of } \mathrm{Quot}(A/\mathfrak{p})\}.$$

Definition 5.2. *Let $v : K \to \Gamma \cup \{0\}$ be a valuation, we define,*

$$R_v := \{x \in K \mid v(x) \leq 1\}.$$

Proposition 5.2. *We have a following correspondence.*

$$\mathrm{Spv}(A) \underset{(v:A \to (\mathrm{Frac}(A/\mathfrak{p})^\times/R^\times) \cup \{0\}/\sim) \leftarrow\shortmid (\mathfrak{p}, R)}{\overset{(v/\sim) \mapsto (\mathrm{supp}(v), R_v)}{\rightleftarrows}} V_A$$

Definition 5.3.

$$v_{\mathfrak{p},R} : A \mapsto (\mathrm{Frac}(A/\mathfrak{p})^\times/R^\times) \cup \{0\}.$$

Corollary 5.1. *In the correspondence above, we have,*

$$\begin{aligned}\mathcal{R}(f_0|f_1, \cdots, f_m) &\mapsto \widetilde{\mathcal{R}}(f_0|f_1, \cdots, f_m),\\ &:= \{(\mathfrak{p}, R) \mid f_0 \notin \mathfrak{p}, \forall i = 1, \cdots, m, f_i \bmod \mathfrak{p} \in (f_0 \bmod \mathfrak{p})_{A/\mathfrak{p}}\}.\end{aligned}$$

Problem 5.8. *Let $\mathcal{U}$ be an ultrafilter on* $\mathrm{Spv}(A)$. *Put,*

$$\mathfrak{p}_{\mathcal{U}} := \{a \in A \mid \{(\mathfrak{p}, R) \in V_A \mid a \in \mathfrak{p}\} \in \mathcal{U}\}.$$

Recall from the lecture note that this is a generic limit of the ultrafilter $\mathcal{U}$ under the direct image given by,

$$V_A \ni (\mathfrak{p}, R) \mapsto \mathfrak{p} \in \mathrm{Spec}(A).$$

Now put,

$$R_{\mathcal{U}} := \{x \in A_{\mathfrak{p}_{\mathcal{U}}} \mid \{(\mathfrak{p}, R) \mid v_{\mathfrak{p},R}(x) \leq 1\} \in \mathcal{U}\}.$$

Show that $(\mathfrak{p}_{\mathcal{U}}, R_{\mathcal{U}})$ is a generic limit of $\mathcal{U}$ in V_A. Further show that $\widetilde{\mathcal{R}}(f_0|f_1, \cdots, f_m)$ is closed under generic limits.

Solution. Let $\Omega = \widetilde{\mathcal{R}}(f_0|f_1, \cdots, f_m)$. In order to apply Lemma 1.3.2, we will prove that

$$g := (\mathfrak{p}_{\mathcal{U}}, R_{\mathcal{U}}) \in \Omega \iff \Omega \in \mathcal{U}.$$

For $\Rightarrow$, let us first examine what $g \in \Omega$ means. Unpacking the definition and under the correspondence, we get

$$v\left(\frac{f_i}{f_0}\right) \leq 1 \Rightarrow \frac{f_i}{f_0} \in \mathbb{R}_v,$$

thus

$$g \in \Omega \iff f_0 \in \mathfrak{p}_{\mathcal{U}}, \forall i = 1, \cdots, m, \frac{f_i}{f_0} \in R_{\mathcal{U}}.$$

Thus we have $D(f_0) \in \mathcal{U}$, and by the construction,

$$E_i := \{(\mathfrak{p}, R) \mid v_{\mathfrak{p},R}(f_i) \leq v_{\mathfrak{p},R}(f_0)\} \in \mathcal{U}, (i = 1, \cdots, m).$$

Since $\mathcal{U}$ is closed under intersection, we obtain

$$\widetilde{\mathcal{R}}(f_0|f_1, \cdots, f_m) = D(f_0) \cap E_1 \cap \cdots \cap E_m \in \mathcal{U}.$$

$\Leftarrow$, inheritting the above notions, we see that

$$\Omega \subseteq D(f_0), E_1, \cdots, E_m.$$

Therefore $D(f_0), E_1, \cdots, E_m \in \mathcal{U}$. □

Definition 5.4. *Let $(X, \mathscr{O}_X)$ be a topological space and $U \in \mathscr{O}_X$. A U-sieve is a subset $\mathscr{S} \subseteq \mathscr{O}_U$ such that*

$$V \in \mathscr{S} \Rightarrow \mathscr{O}_V \subseteq \mathscr{S}.$$

We set $\mathscr{S}|_V := \mathscr{S} \cap \mathscr{O}_V$.

Definition 5.5. *A collection of sieves $\mathcal{T}$ is a Grothendieck topology on X if*

GT-Triv : $\forall U \in \mathscr{O}_X, \mathscr{O}_U \in \mathcal{T}$

GT-Trans : $\forall U \in \mathscr{O}_X$ *and* $V \in \mathscr{O}_U, \mathscr{S} \in \mathcal{T}_U \to \mathscr{S}|_V \in \mathcal{T}_V$

GT-Loc : Let $\mathscr{S} \in \mathcal{T}_U$ and $\mathscr{T}$ be a U-sieve such that for all $V \in \mathscr{S}$, $\mathscr{T}|_V \in \mathcal{T}_V$. Then $\mathscr{T} \in \mathcal{T}_U$.

Problem 5.9. *Let $\mathcal{B}$ be a topology base for X such that for any $\Omega, \Theta \in \mathcal{B}$, $\Omega \cap \Theta$ is a finite union of elements from $\mathcal{B}$. Set,*

$$\mathcal{B}_U = \mathcal{B} \cap \mathscr{O}_U.$$

Let $\mathcal{T}$ be a collection of sieves such that for $U \in \mathscr{O}_X$, $\mathscr{S} \in \mathcal{T}_U$ is a U-sieve such that

$$\forall \Omega \in \mathcal{B}_U, \exists \Theta_1, \cdots, \Theta_n \in \mathscr{S} \cap \mathcal{B}_U, \Omega = \bigcup_{i=1}^{n} \Theta_i.$$

Show that $\mathcal{T}$ is a Grothendieck topology. Furthermore, show that

$$\forall U \in \mathscr{O}_X, \mathscr{S} \in \mathcal{T}_U, U = \bigcup_{V \in \mathscr{S}} V.$$

It seems that finite union assumption is not needed.

Solution. GT-Triv is obvious. For GT-Trans, note that

$$\Omega \subseteq V \subseteq U \Rightarrow \mathscr{S}|_V \cap \mathcal{B}_\Omega = \mathscr{S} \cap \mathcal{B}_\Omega.$$

Thus it remains to show that it satisfies GT-Loc. Let $\mathscr{S}, \mathscr{T}$ be U-sieves such that $\mathscr{S} \in \mathcal{T}_U$. Suppose for any $V \in \mathscr{S}$, we have $\mathscr{T}|_V \in \mathcal{T}_V$. Then for $\Omega \in \mathcal{B}_U$, there are $\Omega_1, \cdots, \Omega_n \in \mathscr{S} \cap \mathcal{B}_\Omega$ such that

$$\Omega = \bigcup_{i=1}^{n} \Omega_i.$$

For each Ω_i, there are $\Theta^i_1, \cdots, \Theta^i_{k_i} \in \mathscr{S} \cap \mathcal{B}_{\Omega_i}$ such that

$$\Omega_i = \bigcup_{j=1}^{k_i} \Theta^i_j.$$

Taking union of these, we conclude $\mathscr{T} \in \mathcal{T}_U$.

It is clear that by construction,

$$U \supseteq \bigcup_{V \in \mathscr{S}} V$$

for U-sieve $\mathscr{S}$. Let $x \in U$ then there is $x \in \Omega \in \mathcal{B}_U$. By the assumption,

$$\Omega = \bigcup \Theta_i$$

for some $\Theta_1, \cdots, \Theta_n \in \mathscr{S} \cap \mathcal{B}_\Omega$. □

Proposition 5.3. *Let L/K be a finite separable extension of fields. Then there is $\alpha \in L$ such that $L = K(\alpha)$.*

Problem 5.10. *Let K be a field equipped with an ultrametric absolute value $|\cdot|_K$. Let $\mathbb{K}$ be a completion of K with respect to $d(x,y) = |x-y|_K$. Show that any finite Galois extension $\mathbb{L}/\mathbb{K}$, there is a dense subfield L such that L/K is finite Galois.*

Solution. By Proposition 5.3, there is $\alpha \in \mathbb{L}$ such that $\mathbb{L} = \mathbb{K}(\alpha)$ Let $f(T) \in \mathbb{K}[T]$ be the minimal polynomial of α. Let $\alpha = \alpha_1, \cdots, \alpha_n$ be the roots of f. Set,

$$\varepsilon = \min_{i<j} |\alpha_i - \alpha_j|_K.$$

Then by the density of $K[T]$ in $\mathbb{K}[T]$, there is g sufficiently close to f of degree n so that its roots $\beta_1, \cdots, \beta_n$ we have,

$$|\beta_i - \alpha_i|_K < \varepsilon.$$

For $\beta := \beta_1$, we may apply Problem 4.5 that $\alpha \in \mathbb{K}(\beta)$. Consider

$$L = K(\beta_1, \cdots, \beta_n).$$

This is Galois and furthermore, if we denote $\widehat{L}$ to be its completion, we get,

$$\mathbb{L} \subseteq \mathbb{K}(\beta) \subseteq \hat{L}.$$

Therefore L is dense. □

Problem 5.11. *Let X be a topological space and $A = C^0(X, K)$ be a ring of continuous functions. Then the map,*

$$\Phi :== [X \ni x \mapsto v_x(f) = [A \ni f \mapsto |f(x)|_K] \in \mathrm{Spv}(A)]$$

is continuous.

Solution. Let $\Omega = \mathcal{R}(f_0|f_1, \cdots, f_m) \subseteq \mathrm{Spv}(A)$ be a rational open

$$D(f_0) = X \backslash f^{-1}(\{0\}).$$

Then we have,

$$\Phi^{-1}(\Omega) = D(f_0) \bigcap_{i=1}^{m} \{x \in D(f_0) \mid |f_i(x)|_K \leq |f_0(x)|_K\}.$$

Since K° is open in K, we have

$$\left|\frac{f_i}{f_0}\right|_K^{-1} (K^\circ) = \{x \in D(f_0) \mid |f_i(x)|_K \leq |f_0(x)|_K\}$$

is open. Therefore Φ is continuous. □

Definition 5.6.

$$\mathrm{Sp}(A) := \{\mathfrak{m} \in \mathrm{Spec}(A) \mid \mathfrak{m} \textit{ is maximal}\}.$$

Proposition 5.4. *For $\mathfrak{m} \in \mathrm{Sp}(\mathbb{T}_n)$, there is a unique extension of $|\cdot|_K$ to $\mathfrak{k}(\mathfrak{m})$ denoted by $|\cdot|_\mathfrak{m}$.*

Definition 5.7. *For $f \in \mathbb{T}_n$, denote $f(\mathfrak{m}) = f \bmod \mathfrak{m}$. Note that*

$$K \hookrightarrow \mathbb{T}_n \to \mathbb{T}_n/\mathfrak{m}.$$

Thus it is indeed a field extension. Consider,

$$\mathrm{Sp}(\mathbb{T}_n) \ni \mathfrak{m} \mapsto ([\mathbb{T}_n \ni f \mapsto |f(\mathfrak{m})|_\mathfrak{m}]/\sim) \in \mathrm{Spv}(\mathbb{T}_n).$$

We will equip $\mathrm{Sp}(\mathbb{T}_n)$ with the coarsest topology making the above map continuous. That is the topology generated by the basis elements,

$$\mathcal{R}_{\mathrm{Sp}(\mathbb{T}_n)}(f_0|f_1, \cdots, f_m) := \{\mathfrak{m} \in \mathrm{Sp}(\mathbb{T}_n) \mid f_0(\mathfrak{m}) \neq 0, \forall i = 1, \cdots, m, |f_i(\mathfrak{m})|_\mathfrak{m} \leq |f_0(\mathfrak{m})|_\mathfrak{m}\}$$

The basis consists of these open sets is denoted by $\mathfrak{Rat}_{\mathrm{Sp}(\mathbb{T}_n)}$.

Proposition 5.5. *$\mathfrak{k}(\mathfrak{m}) = \mathrm{Frac}(\mathbb{T}_n/\mathfrak{m}) = \mathbb{T}_n/\mathfrak{m}$ is a finite field extension of K.*

Proof. We now that $\mathfrak{m}$ is given by $\operatorname{im} \mathrm{ev}_\alpha$ for some $\alpha \in (L^\circ)^m$ where L/K is a finite field extension. Thus,

$$\mathbb{T}_n/\mathfrak{m} = K[\![\alpha_1, \cdots, \alpha]\!].$$

But $\alpha_1, \cdots, \alpha_n$ are of finite order as L/K is finite. We conclude,

$$\mathbb{T}_n/\mathfrak{m} = K[\alpha_1, \cdots, \alpha_n]$$

which is a finite extension of K. □

Problem 5.12. *Show that indeed $\mathfrak{Rat}_{\mathrm{Sp}(\mathbb{T}_n)}$ is the topology base for $\mathrm{Sp}(\mathbb{T}_n)$ making the map continuous.*

Solution. Consider $\mathcal{R}(f_0|f_1, \cdots, f_m) \subseteq \mathrm{Spv}(\mathbb{T}_n)$. Then the preimage is clearly,

$$\mathcal{R}_{\mathrm{Sp}(\mathbb{T}_n)}(f_0|f_1, \cdots, f_m).$$

Therefore, these sets muset be open in $\mathrm{Sp}(\mathbb{T}_n)$. Remains to show that taking a subset of ones that $(f_0, \cdots, f_m) = \mathbb{T}_n$ gives a topology basis.

To do so consider,

$$U = \mathcal{R}_{\mathrm{Sp}(\mathbb{T}_n)}(f_0|f_1, \cdots, f_m).$$

For $x \in U$, we have $f_0(x) \neq 0$, therefore using $|\cdot|_K$ is discrete, we find $a \in K$ such that

$$|a|_K < |f_0(a)|_K.$$

We can take a to be a unit, then the ideal will be the whole and

$$x \in \mathcal{R}_{\mathrm{Sp}(\mathbb{T}_n)}(f_0|f_1, \cdots, f_m, a) \subseteq \mathcal{R}_{\mathrm{Sp}(\mathbb{T}_n)}(f_0|f_1, \cdots, f_m).$$

□

Problem 5.13. *Show that* $\mathfrak{Rat}_{\mathbb{T}_n}$ *is closed under finite intersection in* $\mathrm{Sp}(\mathbb{T}_n)$.

Solution. Clearly,

$$\mathcal{R}_{\mathrm{Sp}(\mathbb{T}_n)}(f_0|f_1,\cdots,f_m)\cap\mathcal{R}_{\mathrm{Sp}(\mathbb{T}_n)}(g_0|g_1,\cdots,g_k)=\mathcal{R}_{\mathrm{Sp}(\mathbb{T}_n)}(f_0g_0|f_ig_j)_{\substack{0\le i\le m\\0\le j\le k}}.$$

As $IJ=(f_ig_j)$ but this is independent of the choice of generators thus it is the whole. □

Problem 5.14. *If* K *is algebraically closed then we have a bijection,*

$$K^{\circ n}\ni x\mapsto\mathfrak{m}_x:=\{f\in\mathbb{T}_n\mid f(x)=0\}.$$

Show that we have a homeomorphism between $K^{\circ n}$ *with topology given by the metric induced by the norm*

$$\|(x_i)_{i=1,\cdots,n}|K^n\|=\max_{i=1,\cdots,n}|x_i|_K,$$

and $\mathrm{Sp}(\mathbb{T}_n)$ *where the topology is given in the previous problem.*

Solution. We first check that the bijection is continuous. Let us denote the bijection by ϕ, take

$$x\in\phi^{-1}(\mathcal{R}_{\mathrm{Sp}(\mathbb{T}_n)}(f_0|f_1,\cdots,f_m)).$$

Then the condition $|f_0(x)|\neq 0$ and $|f_i(x)|\le|f_0(x)|$ is preserved under pertubations (small changes) in x since $f_0,\cdots,f_m$ are all continuous. Thus ϕ is continuous.

Let $B=B(x,\varepsilon)$ where $\varepsilon=|a|_K$ for some $a\in K^\times$. Then

$$\Omega=\mathcal{R}_{\mathrm{Sp}(\mathbb{T}_n)}(f_0|f_1,\cdots,f_n)$$

where $f_0=1$ and

$$f_i=\frac{T_i-x_i}{a}.$$

Then clearly $\phi(B)=\Omega$. Thus ϕ is open. □

Sheet6. G_+-space

Definition 6.1. *Let* $\{U_i\}_{i\in I}$ *be a collection of open basis subsets of a space* X*, we define,*

$$[\{U_i\}_{i\in I}]:=\{V\in\mathscr{O}_X\mid\exists i\in I,V\subseteq U_i\}.$$

Definition 6.2. *A* T_0*-space* X *with a Grothendieck topology* $\mathcal{T}$ *is* G_+ *if every admissible sieve is a covering. A* G_+*-base is a topology base* $\mathcal{B}$ *such that* $[\mathcal{B}_U]\vdash U$ *for any* $U\in\mathscr{O}_X$.

Problem 6.1. *Let* X *be a* T_0*-space and* $\mathcal{B}$ *be a topology base for* X *such that* $\Omega,\Theta\in\mathcal{B}$*,* $\Omega\cap\Theta$ *is a finite union of elements from* $\mathcal{B}$*. Let* $\mathcal{T}$ *be such that for each* $U\in\mathscr{O}_X$*,*

$$\mathcal{T}_U:=\left\{\mathscr{S}\ U\text{-}sieve\middle|\forall\Omega\in\mathcal{B}_U\exists\Theta_1,\cdots,\Theta_n\in\mathcal{B}_\Omega\cap\mathscr{S},\Omega=\bigcup_{i=1}^n\Theta_i\right\}.$$

Show that with this Grothendieck topology, X *becomes* G_+*-space, elements of* $\mathcal{B}$ *are quasi-compact, and it is a* G_+*-base.*

Problem 6.2. *Let* X *be a* T_0 *space and* $\mathcal{B}$ *be a topology base from Problem 9 from the previous sheet. Show that the Grothendieck topology in that problem,* X *is* G_+*-space and* $\mathcal{B}$ *is a* G_+*-base consisting of* G_+*-quasi-compact sets.*

Solution. From previous problem, X being G_+-space is immediate under the assumption that X is T_0.

To show that $\mathcal{B}$ is a G_+ base, we must verify that $[\mathcal{B}_U] \vDash U$. Let $\Omega \in \mathcal{B}_U$, then as $\Omega \in \mathcal{B}_\Omega$ and $\Omega \subseteq U$, $\Omega \in \mathcal{B}_\Omega \cap [\mathcal{B}_U]$, thus $\Theta_1 = \Omega$ can be taken.

Recall that for $\mathscr{S} \in \mathcal{T}_\Omega$, we have finitely many $\Theta_1, \cdots, \Theta_n \in \mathscr{S} \cap \mathcal{B}_\Omega$ such that

$$\Omega = \bigcup_{i=1}^{n} \Theta_i.$$

Therefore, $[\Theta_1, \cdots, \Theta_n] \vDash \Omega$. Indeed, for any $\Xi \in \mathcal{B}_\Omega$, we have

$$\Xi \cap \Theta_i = \bigcup \Xi_j^i.$$

□

Problem 6.3. *Let X be a G_+-space and $\mathcal{B}$ be a G_+-base containing X. Recall that a van der Put point is a subset $\xi \subseteq \mathcal{B}$ such that*

(V1) $X \in \xi$,

(V2) $\Omega, \Theta \in \xi \Rightarrow \Omega \cap \Theta \in \xi$,

(V3) $\Omega \subseteq \Theta$ *and* $\Omega \in \xi$ *then* $\Theta \in \xi$,

(V4) for any covering sieve $\mathscr{S} \subseteq \mathscr{O}_\Omega$, $\mathscr{S} \cap \xi \neq \emptyset$.

The set of these subsets is denoted by $X^*_{\mathcal{B}}$ and in particular, we write $X^*_{\mathscr{O}_X} = X^*$.

Show that we have a bijection,

$$X^* \ni \xi \mapsto \xi \cap \mathcal{B} \in X^*_{\mathcal{B}}.$$

The inverse is given by

$$\xi_{\mathcal{B}} \mapsto \{U \in \mathscr{O}_X \mid \exists \Omega \in \xi_{\mathcal{B}}, \Omega \subseteq U\}.$$

Solution. The map $X^* \to X^*_{\mathcal{B}}$ is well-defined. Since, $X \in \xi \cap \mathcal{B}$ and $\mathcal{B} \subseteq \mathscr{O}_X$. Thus first three conditions are immediate. For the fourth, note that for each $\Omega \in \mathcal{B}$, and Ω-sieve, $\mathscr{S}$ we have $U \in \mathscr{S} \cap \xi \neq \emptyset$. Using G_+-base property, we have $V \in [\mathcal{B}_U] \cap \xi \neq \emptyset$. Thus for some $\Theta \in \mathcal{B}_U$, we have

$$V \subseteq \Theta \subseteq U.$$

As $V \in \xi, \Theta \in \xi$. And as $\mathscr{S}$ is a sieve and $U \in \mathscr{S}, \Theta \in \mathscr{S}$. We conclude $\Theta \in \mathscr{S} \cap \xi_{\mathcal{B}}$.

For the other direction, let $\xi_{\mathcal{B}} \in X^*_{\mathcal{B}}$ and ξ be the image. We have, $X \in \xi$ and for $\Omega \subseteq U, \Theta \subseteq V \in \xi$, we have $\Omega \cap \Theta \subseteq U \cap V \in \xi$. If $\Omega \subseteq V \subseteq U$ then $\Omega \subseteq U \in \xi$. The fourth is clear as $\xi_{\mathcal{B}} \subseteq \xi$.

To prove these are mutual inverses, $X^*_{\mathcal{B}} \to X^* \to X^*_{\mathcal{B}}$ is clear. For the other direction, use the property of topology basis. □

Problem 6.4. *In the setting of Problem 6.1, the condition (V4) can be rephrased as,*

$$\forall \Omega \in \xi_{\mathcal{B}}, [\exists \Theta_1, \cdots, \Theta_n \in \mathcal{B} \text{ s. t. } \Omega = \bigcup_{i=1}^{n} \Theta_i \Rightarrow \exists i = 1, \cdots, n, \Theta_i \in \xi_{\mathcal{B}}]$$

Solution. If $\Omega = \bigcup_{i=1}^{n} \Theta_i$, then $\mathcal{S} := [\Theta_1, \cdots, \Theta_n] \vDash \Omega$, therefore $\mathcal{S} \cap \xi_{\mathcal{B}} \neq \emptyset$. Thus $\Theta_i \in \xi_{\mathcal{B}}$ for some i.

For the other direction, note that $\mathcal{B}$ consits of G_+-quasi-compact elements under the Grothendieck topology forcing quasi-compactness. Therefore, for $\mathscr{S}$ an Ω-sieve, we obtain a subsieve,

$$[\Theta_1, \cdots, \Theta_n] \vDash \Omega.$$

Therefore $\Theta_i \in \xi_{\mathcal{B}}$ and $\Theta_i \in \mathscr{S}$. □

Problem 6.5. *For $R \in (0,1]_{\mathbb{R}}$, show that*

$$\mathbb{T}_1 \ni f = \sum_{i=0}^{\infty} f_i T^i \overset{N_R}{\mapsto} \max_{i \in \mathbb{Z}_{\geq 0}} \{R^i |f_i|_K\}$$

defines a multiplicative norm.

Solution. We have,

$$fg = \sum_{i=0}^{\infty} \left(\sum_{j=0}^{i} f_j g_{i-j} \right) T^i.$$

Therefore,

$$R^i \left| \sum_{j=0}^{i} f_j g_{i-j} \right| = R^i |f_{j_0} g_{i-j_0}| \leq R^{j_0} |f_{j_0}| R^{i-j_0} |g_{i-j_0}| \leq N_R(f) N_R(g).$$

This is attainable. □

An important observation is when K is algebraically closed, we have $(0,\infty)/|K^\times|$ is a torsion free group with identity 1. Since if R is a torsion, this is a solution for the equation $X^m = 0$. This leads to the notion that for $R \in (0,1)_{\mathbb{R}} \backslash |K^\times|$, we define,

$$\mathbb{T}_1 \backslash \{0\} \ni f \mapsto N_R(f) = \nu \in |K^\times| R^{\mathbb{Z}}.$$

We denote the R-power part of ν by j_ν.

Problem 6.6. *Let K be algebraically closed and $R \in (0,1)_{\mathbb{R}} \backslash |K^\times|$. Denote $\nu = N_R(f)$ and $j = j_\nu$. Show that there are $r_- \in (0,R)_{\mathbb{R}}$ and $r_+ \in (R,\infty)_{\mathbb{R}}$ such that for all $z \in K$*

$$z \in [r_-, r_+]_{\mathbb{R}} \Rightarrow |f(z)| = \nu \left(\frac{|z|}{R} \right)^j.$$

Solution. We have,

$$N_R(f) = \nu = R^j |f_j|_K.$$

We can also write,

$$|f(z)| = \max_{i \in \mathbb{Z}_{\geq 0}} |f_i| |z|^i.$$

So what we want is

$$|f_i||z|^i < |f_j||z|^j = \frac{\nu}{R^j} |z|^j \iff |f_i||z|^{i-j} < \frac{\nu}{R^j}.$$

For $i < j$, we take,

$$|z|^{j-i} > \frac{R^j}{\nu} |f_i|.$$

Thus take $z_- = \max_{0 \leq i < j} \sqrt[j-i]{\frac{R^j}{\nu} |f_i|}$. On the other hand, for $i > j$,

$$|f_i||z|^{i-j} < \frac{\nu}{R^j} \iff |z|^{i-j} < \frac{\nu}{R^j} \frac{1}{|f_i|}.$$

Since $(f_n)_{n \in \mathbb{N}}$ is bounded, we can take an infimum r_+. □

Problem 6.7. *Suppose K is algebraically closed thus we identify K° with $\mathrm{Sp}(\mathbb{T}_1)$. Take $R \in (0,1)_{\mathbb{R}} \backslash |K^\times|$ and a rational open*

$$\Omega = \mathcal{R}_{\mathrm{Sp}(\mathbb{T}_1)}(f_0 | f_1, \cdots, f_m).$$

Show the following statements.

1. *If for each $i = 1, \cdots, m$, we have $N_R(f_i) \leq N_R(f_0)$ then there are $r_- \in (0, R), r_+ \in (R, 1)$ such that $D(0, r_+) \subseteq \Omega \cup D(0, r_-)$.*

2. *If there is $1 \leq i \leq m$ such that $N_R(f_i) > N_R(f_0)$, then there are $r_- \in (0, R), r_+ \in (R, 1)$ such that $\Omega \cap D(0, r_+) = D(0, r_-)$.*

Solution. For each $i = 0, \cdots, m$, take r_-^i, r_+^i as in Problem 5 and denote the power of R by j_i. Then,

$$\frac{|f_i(z)|}{|f_0(z)|} \leq \frac{\nu_i}{\nu_0} \left(\frac{|z|}{R}\right)^{j_i - j_0} \leq 1.$$

This holds in,

$$I = \bigcap_{i=0}^{n} (r_-^i, r_+^i).$$

Thus take $(r_-, r_+) \subseteq I$ this is possible as all interval contains R as an internal point. Therefore, we have,

$$r_- \leq |z| \leq r_+ \Rightarrow z \in \Omega.$$

In particular $D(0, r_+) \subseteq \Omega \cup D(0, r_-)$.

Suppose there is j such that $N_R(f_j) > N_R(f_0)$. Then for some $|z| \in [r_-, r_+]$ we have,

$$|f_j(z)| = N_R(f_j) > N_R(f_0) = |f_0(z)|.$$

Thus we get,

$$D(0, r_+) \backslash D(0, r_-) \cap \Omega = \emptyset.$$

By taking a union with $D(0, r_-)$ both sides, we get the desired equality. □

Sheet7. Topological Modules

Problem 7.1. *Suppose K is algebraically closed thus we identify K° with $\mathrm{Sp}(\mathbb{T}_1)$. Take $R \in (0, 1)_{\mathbb{R}} \backslash |K^\times|$ and a rational open Show that there is a van der Put point ξ of $\mathrm{Sp}(\mathbb{T}_1)$ such that a rational open set $\Omega = \mathcal{R}(f_0 | f_1, \cdots, f_m)$ belongs to ξ if and only if one of the first statement of the following holds.*

1. *$N_R(f_j) \leq N_R(f_0)$ for $1 \leq j \leq n$ or equivalently there are $r_- \in (0, R)$ and $r_+ \in (R, 1)$ such that $D(0, r_+) \subseteq \Omega \cup D(0, r_-)$.*

2. *There is j such that $N_R(f_j) > N_R(f_0)$ or equivalently there are $r_- \in (0, R)$ and $r_+ \in (R, 1)$ such that $\Omega \cap D(0, r_+) \subseteq D(0, r_-)$.*

Solution. Let $\xi_{\mathfrak{Rat}} \subseteq \mathfrak{Rat}_{\mathbb{T}_n}$ be the subset of rational opens satisfying the first condition. Then as $X = \mathcal{R}(1|)$ satisfies it we have $X \in \xi_{\mathfrak{Rat}}$. For the intersection $\Omega \cap \Theta$, we can take smaller r_+ and bigger r_- from corresponding values for Ω, Θ to satisfy the inclusion. For the upward closure, $D(0, r_+) \subseteq \Omega \cup D(0, r_-)$ and $\Omega \subseteq \Theta$ then $D(0, r_+) \subseteq \Theta \cup D(0, r_-)$. For the fourth condition, recall that we equipped $\mathrm{Sp}(\mathbb{T}_n)$ with a Grothendieck topology enforcing quasi-compactness to $\mathfrak{Rat}_{\mathbb{T}_n}$. Thus we need to show if

$$\Omega = \bigcup_{i=1}^{n} \Theta_i,$$

and $\Omega \in \xi_{\mathfrak{Rat}}$, then there is j such that $\Theta_j \in \xi_{\mathfrak{Rat}}$. Suppose not, thus all Θ_i satisfy the second condition. Then Ω must also satisfy it which is a contradiction.

By the bijection of van der Put points, we obtain that there is a unique $\xi \subseteq \mathscr{O}_{\mathrm{Sp}(\mathbb{T}_n)}$ such that $\xi_{\mathfrak{Rat}} = \xi \cap \mathfrak{Rat}_{\mathbb{T}_n}$. □

Problem 7.2. *Show that for $x \in K^\circ$ and $R \in (0, 1] \cap K^\times$, $B(x, R) \in \mathfrak{Rat}_{\mathbb{T}_1}$.*

Solution. By the assumption, there is $r \in K^\times$ such that $|r|_K = R$. Then,

$$D(x,r) = \mathcal{R}\left(1 \middle| \frac{T_1 - x_1}{r}, \cdots, \frac{T_n - x_n}{r}\right)$$

□

Problem 7.3. *Let v be a van der Put point for $\mathrm{Sp}(\mathbb{T}_1)$. Suppose for any $\rho \in (0,1] \cap |K^\times|$, we have,*

$$D(0,\rho) \in v \iff \rho > R.$$

Then we have $v = \xi$.

Solution. Suppose we have $\Omega \in \xi$, by the non-discreteness of the absolute value, we may pick $\rho_- \in (0,R), \rho_+ \in (R,1)$ such that

$$D(0,\rho_+) \subseteq \Omega \cup D(0,\rho_-).$$

Since $\rho_+ > R$ and by the upward closure, $\Omega \cup D(0,\rho_-) \in v$. Using the alternative characteristic of van der Put points, we have $\Omega \in v$.

On the other hand, if $\Omega \notin \xi$, then

$$\Omega \cap D(0,\rho_+) \subseteq D(0,\rho_-).$$

Since $\mathfrak{Rat}_{\mathbb{T}_n}$ is closed under finite intersections and thus $\Omega \in v$ would imply $D(0,\rho_-) \in v$ which is a contradiction.

By the bijection of van der Put points between topology bases, we obtain the claim. □

Definition 7.1. *Let $v \in \mathrm{Spv}(A)$ be a valuation, we define,*

$$K_{\leq r}(v) := \{a \in A \mid v(a) \leq r\}, K_{<r}(v) := \{a \in A \mid v(a) < r\}.$$

Problem 7.4. *Set,*

$$\mathfrak{B}_{\leq R} := \{D(x,R) \mid x \in K^\circ\}, \mathfrak{B}_{<R} := \{B(x,R) \mid x \in K^\circ\}.$$

Furthermore, we set,

$$\mathfrak{B}_{\leq} := \bigcup_{R \in (0,1]_{\mathbb{R}}} \mathfrak{B}_{\leq R}, \mathfrak{B}_{<} := \bigcup_{R \in (0,1]_{\mathbb{R}}} \mathfrak{B}_{<R}.$$

With these, we denot $\mathfrak{B} = \mathfrak{B}_{\leq} \cup \mathfrak{B}_{<}$.

Show that for $B_1, B_2 \in \mathfrak{B}$, either one contains the other or they are disjoint.

Solution. Suppose $B_1 = D(x_1,r_1), B_2 = D(x_2,r_2)$ and $r_1 \leq r_2$. Suppose $B_1 \cap B_2 \neq \emptyset$ and take $x \in B_1 \cap B_2$. Then for any $y \in B_1$,

$$|y - x_2| = |y - x + x - x_2| = |y - x_1 + x_1 - x + x - x_2| \leq \max r_1, r_2 = r_2.$$

□

Let A be a topological ring and M be a topological A-module. Take K to be as usual but drop the algebraically closedness.

Problem 7.5. *If $X \subseteq A, Y \subseteq M$ are both bounded then so is XY.*

Solution. Let $U \subseteq M$ be a neighborhood of 0. Then there is a neighborhood V of 0 in A such that $VY \subseteq U$. Since X is bounded, there is another neighborhood W of 0 such that $WX \subseteq V$. Thus $WXY \subseteq U$. □

Problem 7.6. *Let $R = K \oplus K\varepsilon$ where $\varepsilon^2 = 0$. Show that $X = K\varepsilon$ is topologically nilpotent but not powerbounded.*

Solution. □

Problem 7.7. *Suppose $X, Y \subseteq A$ are topologically nilpotent then so is $X \cup Y$.*

Solution. Let $U \subseteq A$ be a neighborhood of 0. Then we can take a neighborhood $\tilde{U}$ such that $\tilde{U}\tilde{U} \subseteq U$ by the continuity of multiplication. Then there are $m, n \in \mathbb{N}$ such that

$$\bigcup_{k \geq m} X^k, \bigcup_{l \geq n} Y^l \subseteq \tilde{U}.$$

Take a bigger m or n, then □

Sheet8. Topological Rings

Problem 8.1. *Let $f : A \to B$ be a morphism of topological rings which is open. Then if $M \subseteq A$ is bounded then so is $f(M)$ in B.*

Solution. Let $U \subseteq B$ be a neighborhood of 0, then $f^{-1}(U)$ is open, thus there is $V \subseteq A$ such that $VM \subseteq f^{-1}(U)$. Since $f(V)$ is open,

$$f(V)f(M) \subseteq U$$

will do. □

Problem 8.2. *Let $I \subseteq A$ be an ideal and equip A/I with the quotient topology. Then the image any bounded set $M \subseteq A$ under the quotinet map is bounded.*

Solution. We aim to use the previous problem. That is we want to show that π is open. Let $U \subseteq A$ be open then,

$$\pi^{-1}(\pi(U)) = \bigcup_{a \in I} U + a.$$

This is open as $U + a$ is open by the continuity of addition thus it is a union of open sets. Therefore, $\pi(U)$ is open by the definition of quotinet topology. □

Problem 8.3. *Let K be a field as usual and $A = K \oplus K\varepsilon$ where $\varepsilon^2 = 0$. Equip the topology of K^2 to A as norm space. Set,*

$$X = K\varepsilon, Y = \{t \cdot x \mid t \in K, |t| \leq r\}$$

for some $r \in (0, 1)$. Show that Y is topologically nilpotent but $X \cup Y$ is not.

Solution. Clearly Y is topologically nilpotent as any neighborhood of 0 contains $B(0, r^m)$ for some m. But clearly $X \subseteq (X \cup Y)^m$ for any m. No open set of the form $B(0, t)$ for $t \in (0, 1)$ contains X. □

Definition 8.1. *Let A be a ring and $A^+ \subseteq$, we set,*

$$\mathrm{Spa}(A, A^+) = \{v \in \mathrm{Cont}(A) \mid \forall a \in A^+, v(a) \leq 1\}.$$

Problem 8.4. *Let $s \in \mathrm{Spa}(A, A^+)$. Then a horizontal specialization of s in $\mathrm{Spv}(A)$ is also in $\mathrm{Spa}(A, A^+)$.*

Solution. We can rephrase the statement that

$$\forall a \in A^+, v(a) \leq 1.$$

to the following,

$$\forall a \in A^+, v \in \mathcal{R}(1|a).$$

Since $\mathrm{Spa}(A, A^+)$ has the strong topology and $\Omega = \mathcal{R}(1|a)$ is a basis open element, we have,

$$s \in \Omega \iff \eta \in \Omega,$$

where η is the specialization. Therefore $\eta \in \mathrm{Spa}(A, A^+)$. □

Problem 8.5. *Let A be a topological ring and X be either $\mathrm{Cont}(A)$ or $\mathrm{Spa}(A, A^+)$. Show that $x \in X$ is a closed point if and only if $\Gamma_x = c\Gamma_x$.*

Solution. Suppose $\Gamma_x \neq c\Gamma_x$. Then $x|_{c\Gamma_x}$ is a proper specialization and belongs to X, thus contradicts to that $\{x\}$ is closed. □

Problem 8.6. *Suppose A is a topological ring and X as same in the previous problem. Show that if A admits a topologically nilpotent unit then every point in X is closed.*

Solution. Let $v \in X$ and $a \in A$ be a topologically nilpotent unit. Then $v(a) < 1$. Then we also have $v(a^{-1}) > 1$. Note that for any $\gamma \in \Gamma_v$, there is $n \in \mathbb{N}$ such that

$$v(a^n) < \gamma < v(a^{-n}).$$

Thus $\gamma \in c\Gamma_v$. □

Problem 8.7. *Suppose X is either $\mathrm{Cont}(A)$ or $\mathrm{Spa}(A, A^+)$. Let $\eta \in X$. Set s be a specialization of η in $\mathrm{Spv}(A)$ and x be a specialization of η in X such that s is a specialization of x in $\mathrm{Spv}(A)$. Show that in this case, we have, $x \in X$ and*

$$\eta \rightsquigarrow x \rightsquigarrow s$$

hodsl in X.

Solution. Since we have this specialization relation in $\mathrm{Spv}(A)$, we obtain,

$$\mathrm{supp}(\eta) \subseteq \mathrm{supp}(x) \subseteq \mathrm{supp}(s).$$

Therefore, combined with usual conditions of specialization, we have for $a \in A \setminus \mathrm{supp}(x)$,

$$K_{s<a} \subseteq K_{x<a} \subseteq K_{\eta<a}.$$

Therefore x is continuous. Since we consider the strong topology, the relation holds in X. □

From the rest of the problems of this sheet, we assume K to be algebraically closed and $X = \mathrm{Sp}\,\mathbb{T}_1$.

Problem 8.8. *Let $\xi \in X^*$. For $R \in (0, 1]$, show that $\xi \cap \mathfrak{B}_{<R}, \xi \cap \mathfrak{B}_{\leq R}$ have at most one element.*

Solution. Let $B_1 = B(x_1, R), B_2 = B(x_2, R)$ and $B_1, B_2 \in \xi \cap \mathfrak{B}_{<R}$. From the previous problem we know that either $B_1 = B_2$ or $B_1 \cap B_2 = \emptyset$. Suppose $B_1 \cap B_2 = \emptyset$. By the forth condition of van der Put points, and that $\emptyset$ covers $\emptyset$ in G_+-space, this would contradict as $\xi \cap \emptyset = \emptyset$. Therefore only $B_1 = B_2$ is possible. Others can be shown similarly. □

This leads us to the notion of $D(\xi, R), B(\xi, R)$ if they exist.

$$M_\xi := \{R \in (0, 1] \mid \xi \cap \mathfrak{B}_{\leq R} \neq \emptyset\}.$$

Problem 8.9. *Show that $1 \in M_\xi$ and $D(\xi, 1) = K^\circ$.*

Solution. We have $K^\circ = D(0, 1)$ which is identified with the whole space as K is algebraically closed. Therefore $1 \in M_\xi$ and in particular $D(\xi, 1) = D(0, 1) = K^\circ$. □

Notation 8.1. $R_\xi = \inf M_\xi$.

Problem 8.10. *Let $\Omega = \mathcal{R}(f_0 | f_1, \cdots, f_m) \in \mathfrak{Rat}_X$. Show that there is $\varepsilon \in (0, \infty)$ such that $\forall x \in \Omega, |f_0(x)| \geq \varepsilon$.*

Solution. Since $(f_0, f_1, \cdots, f_m) = \mathbb{T}_1$, and $|f_0(x)| \geq |f_i(0)|$ for $x \in \Omega$, we have $f_0(x) \neq 0$. More explicitly, there are $g_0, \cdots, g_m \in \mathbb{T}_1$ such that

$$1 = \sum_{i=0}^{n} g_i f_i.$$

By using the ultrametric property, we have,

$$\begin{aligned} 1 &\leq \max_{i=0,\cdots,m} |g_i(x) f_i(x)|, \\ &\leq |f_0(x)| \max_{i=0,\cdots,m} |g_i(x)|. \end{aligned}$$

Therefore, set $C = \max_{i=0,\cdots,m} |g_i(x)|$. Then we obtain,

$$\forall x \in \Omega, C^{-1} \leq |f_0(x)|.$$

□

Problem 8.11. *Show that in the above problem, we have Ω is clopen in an ordinary topology in X.*

Solution. For each $i = 1, \cdots, m$, we set,

$$F_i(x) = |f_i(x)| - |f_0(x)|.$$

This function takes only positive values on $X \backslash \Omega$. We then see immediately that

$$X \backslash \Omega = \bigcup_{i=1}^{m} F_i^{-1}((0, \infty)).$$

□

Problem 8.12. *If $R_\xi = 0$, then there is $x \in X$ such that*

$$\xi = \{U \in \mathscr{O}_X \mid x \in U\}.$$

Solution. By the bijection of van der Put points, we only need to check that

$$\xi \cap \mathfrak{Rat}_X = \{\Omega \in \mathfrak{Rat}_X \mid x \in \Omega\}.$$

Let $(R_n)_{n \in \mathbb{N}}$ be a sequence in M_ξ converging to 0. We may further assume that this is monotonically decreasing. Hence

$$D(\xi, R_n) \subseteq D(\xi, R_{n+1}).$$

We then pick $(x_n)_{n \in \mathbb{N}}$ such that $x_n \in D(\xi, R_n)$. Then

$$|x_n - x_m| \leq R_{\min m,n} \to 0.$$

Thus this is Cauchy. By the completeness, we obtain the limit x. Since $K^\circ = B_1(\xi) \in \xi$, we may construct the sequence so that $x \in K^\circ$.

Suppose $\Omega \in \xi \cap_{\mathfrak{Rat}_X}$. Suppose further that $x \notin \Omega$, then as it is clopen, we find $\Theta \in \mathfrak{Rat}_X$ such that $x \in \Theta$. Using the continuity of polynomials, we may pick R_n such that $D(\xi, R_n) \subseteq \Theta$. However, $B(\xi, R_n) \in \xi$ therefore $B(\xi, R_n) \cap \Omega \neq \emptyset$. This is a contradiction.

Suppose $x \in \Omega$, then we can pick $B(\xi, R_n) \subseteq \Omega$. But $B(\xi, R_n) \in \xi$ and using the upward closure $\Omega \in \xi$. □

Sheet9. Van der Put points

Throughout the sheet, we assume K is algebraically closed, non-discrete and complete with respect to an ultrametric absolute value $|\cdot|_K$. Set $X = \mathrm{Sp}(\mathbb{T}_1)$.

Lemma 9.1. *$D(0, r)$ is open. Therefore $D(x, r)$ is open for any $x \in K^\circ$.*

Proof. Let $x \in D(0, r)$ and if $|x - y| < r$, then

$$|y| = |y - x + x| \leq \max\{r, |x|\} = r.$$

Therefore $B(x, r) \subseteq D(0, r)$. □

Problem 9.1. *Let $\xi \in X^*$ be such that and $R_\xi \in (0, 1) \backslash |K^\times|$ and for any $r \in (R_\xi, 1]$, $D(0, r) \in \xi$.*

Solution. By the lemma, we have,

$$r > R_\xi \Rightarrow D(0, r) \in \xi.$$

Therefore, we have,

$$D(0, \rho) \in \xi \iff \rho \in (0, 1] \cap K^\times.$$

□

Definition 9.1. *Let F be a field with an absoltue value. F is said to be spherically complete if given any set $\mathcal{L}$ of balls in K which is linearly ordered by inclusion admits a non-empty intersection.*

Problem 9.2. *If F is spherically complete then it is complete.*

Solution. Let $(x_n)_{n\in\mathbb{N}}$ be a Cauchy sequence in F such that $|x_n - x_m| \leq r_m$ for $m \leq n$. Set $D_n = D(x_n, r_n)$. Then $\{D_n\}_{n\in\mathbb{N}}$ is linearly ordered by inclusions. Therefore,

$$x \in \bigcap_{n\in\mathbb{N}} D_n.$$

Clearly $x_n \to x$. □

Problem 9.3. *Suppose $|K^\times|$ is a discrete set, then spherically completeness coincides with completeness.*

Solution. Remains to show that completeness implies spherical completeness. Let $\mathcal{L}$ be a linearly ordered set of balls. Set R be the infimum of radii in $\mathcal{L}$.

If $R = 0$, then for $B_1 \supseteq B_2 \supseteq \cdots$ in $\mathcal{L}$ such that the radii go to 0. Set $(x_n)_{n\in\mathbb{N}}$ be such that $x_n \in B_n$. Then by the completeness of F, we find a limit of it which is contained in the intersection.

If $R \neq 0$, then $R = |a|$ for some $a \in K^\times$ by the discreteness. In particular, $B_1 \supseteq B_2 \supseteq \cdots$ stabilizes at some points. Thus the intersection will be the ball after the stabilization which is non-empty. □

Problem 9.4. *If $|F^\times| \subseteq (0,1)$ is dense and F admits a countable dense subset. Then K is not spherically complete.*

Proof. Let $D = \{d_1, d_2, \cdots\}$ be the countable dense set and $r > 0$. As $|K^\times|$ is dense, we find a sequence $(r_n)_{n\in\mathbb{N}}$ in $|K^\times|$ such that $r_n \to r$. Set $B_1 = D(d_1, r_1)$. Suppose B_n is constructed. We aim to construct B_{n+1} such that $d_{n+1} \notin B_{n+1}$. If $d_{n+1} \notin B_{n+1}$, then simply take a smaller ball in B_n with radius r_{n+1}. If $d_{n+1} \in B_n$, then pick $y \in B_n$ such that $|y - d_{n+1}| > r_{n+1}$ and set $B_n = D(y, r_{n+1})$. Then consider,

$$x \in \bigcap_{n\in\mathbb{N}} B_n.$$

Then as D is dense, there is d_N such that $|x - d_N| < r$. Hence $d_N \in B_n$ for any n but this is a contradiction. □

Definition 9.2. *Let F be a field with an absolute value such that $|F^\times|$ is dense in $(0,1)$. A mock circle of radius r is a linearly ordered set $\mathfrak{M}$ of disks in F satisfying the following.*

1. *For all $R \in (r, \infty)$, $\mathfrak{M} \cap \mathfrak{K}_{\leq R} \neq \emptyset$.*
2. *The intersection of all elements in $\mathfrak{M}$ is empty.*

Definition 9.3. *An element $f \in \mathbb{T}_1$ is multiplicatively equivalent to a polynomial p if there is $u \in \mathbb{T}_1^\times$ such that $f = up$.*

Problem 9.5. *Let $f \in \mathbb{T}_1$ and $\mathfrak{M}$ be a mock circle in K such that $K^\circ \in \mathfrak{M}$. Then there is $B \in \mathfrak{M}$ such that $B \ni x \mapsto |f(x)|$ is constant.*

Solution. As K is algebraically closed and every $f \in \mathbb{T}_1$ is multiplicatively equivalent to a polynomial, f admits fnitely many zeros $a_1, \cdots, a_n$. For each i, pick $B_i \in \mathfrak{M}$ such that $a_i \notin B_i$. As $\mathfrak{M}$ is linearly ordered, pick the smallest B. Therefore B has no zeros of f. Therefore f is constant on B by Problem 9.3. □

Since $\mathfrak{M}$ is linearly ordered this leads us to the following definition.

Definition 9.4. *For a mock circle $\mathfrak{M}$ and $f \in \mathbb{T}_1$. We define,*

$$\|f\|_{\mathfrak{M}} = |f(x)|$$

where $x \in B$ and f takes a constant value on B.

Problem 9.6. *Let $\Omega = \mathcal{R}_X(f_0|f_1, \cdots, f_m) \subseteq K^\circ$ be rational open. Show the following.*

1. *If for all $i = 1, \cdots, m$, $\|f_i\|_{\mathfrak{M}} \leq \|f_0\|_{\mathfrak{M}}$, then there is $B \in \mathfrak{M}$ such that $B \subseteq \Omega$.*
2. *If there is $i = 1, \cdots, m$ such that $\|f_i\|_{\mathfrak{M}} > \|f_0\|_{\mathfrak{M}}$, then there is $B \in \mathfrak{M}$ such that $B \cap \Omega = \emptyset$.*

Furthermore, there is a van der Put point $\xi_{\mathfrak{M}} \in X^*$ such that $\xi_{\mathfrak{M}} \cap \mathfrak{Rat}_X$ consists of rational opens satisfying the first alternative.

Solution. If for all $i = 1, \cdots, m$, $\|f_i\|_{\mathfrak{M}} \leq \|f_0\|_{\mathfrak{M}}$, then take a smallest B in $\mathfrak{M}$ such that all f_i are constant on it. Then clearly $B \subseteq \Omega$. Suppose other wise, then again taking the smallest B, we conclude $B \cap \Omega = \emptyset$.

The last statement is to show that such collection defines a van der Put point. Set $\xi = \xi_{\mathfrak{M}} \cap \mathfrak{Rat}_X$, then $X \in \xi$. By the linearly order on $\mathfrak{M}$, it is closed under finite intersection. For the upward closure, clearly $B \subseteq \Omega \subseteq \Theta$ then $B \subseteq \Theta$.

Finally, for the finite covering condition, suppose there is Ω and $\Theta_1, \cdots, \Theta_k$ with $\Omega = \Theta_1 \cup \cdots \cup \Theta_k$. Suppose none of Θ_i satisfies the condition. Then there is B_i for each i from $\mathfrak{M}$ such that $B_i \cap \Theta_i = \emptyset$. Then by choosing the smallest, we obtain there is B_{i_0} with $B_{i_0} \cap \Omega$ a contradiction. □

Problem 9.7. *Let $\xi \in X^*$ be such that $R_\xi \in (0,1)$. Suppose $\bigcap_{r \in (R_\xi, 1]} D(\xi, R) = \emptyset$. Set a mock circle $\mathfrak{M}$ to be the set of disks such that $D(\xi, r)$ for $r \in (R_\xi, 1]$ or $D(0, r)$ for $r \in (0, \infty)$. Show that $\xi = \xi_{\mathfrak{M}}$.*

Solution. $\mathfrak{M}$ is indeed a mock circle. The intersection being emptyset is clear. For linearly orderedness, it is obvious if two disks are the same kind. And it remains to show that for $D(\xi, r)$ and $D(0, s)$ for some $s < 1$. Since $D(0,1) \subseteq D(0, R)$ for all $R \geq 1$, we may assume $R \in (0,1)$. Suppose □

Definition 9.5. *Let $r \in (0,1) \cap K^\times$. For $f \in \mathbb{T}_1$. As before, we define,*

$$\nu_r(f) = \max_{j \in \mathbb{Z}_{\geq 0}} r^j |f_j|.$$

We define $j_r(f)$ to be such that

$$r^{j_r(f)} |f_{j_r(f)}| = \nu_r(f),$$

and it is maximum among them. With these, we define,

$$N_r(f) = \begin{cases} 0, & (f = 0), \\ (\nu_r(f), j_r(f)), & (f \neq 0). \end{cases}$$

The order in $(0, \infty) \times \mathbb{Z}$ is given by the lexicographic order.

Problem 9.8. *Let $f \in \mathbb{T}_1 \backslash \{0\}$ and $N_r(f) = (\nu, j)$. Show that there is $r < R < 1$ such that*

$$\forall x \in K^\circ, r < |x| < R \Rightarrow |f(x)| = \nu \times \left(\frac{|x|}{r} \right)^j.$$

Solution. Recall that inside the absolute value, the largest term is dominant. For all $i < j$, we have,

$$r^i |f_i| \leq r^j |f_j| \Rightarrow \frac{|f_i|}{|f_j|} \leq r^{j-i}.$$

For $r < |x|$, we have,

$$\frac{|f_i||x^i|}{|f_j||x^j|} = \frac{|f_i|}{|f_j|} |x|^{i-j} \leq \left(\frac{r}{x} \right)^{j-i} < 1.$$

Therefore, we obtain,

$$|f_i x^i| < |f_j x^j|.$$

By definition, we have,

$$M = \max\{|f_j| + 1, |f_{j+1}|, \cdots\}.$$

For $j < i$, we have,

$$r^i|f_i| < r^j|f_j|$$

by the maximality of j. Therefore the inequality holds for any $x \in K^\circ$ sufficiently close to r. As f_i-s are bounded, we can take $R > r$. □

Problem 9.9. *Let A be a nat-ring and $X \subseteq A$ be a topologically nilpotent set. Show that the subgroup X' generated by X in A is also topologically nilpotent.*

Solution. Let $U \subseteq A$ be a neighborhood of 0 which is also a subgroup. This is possible as A is a nat-ring. Then there is $N \in \mathbb{N}$ such that $\bigcup_{n \geq N} X^n \in U$. As elements of X^n are product of n-many elements from X and as U is a subgroup, we have $X'^n \subseteq U$ for any $n \geq N$. □

Sheet10. Classifying Valuations

Denote $X = \operatorname{Sp} \mathbb{T}_1$.

Problem 10.1. *Let $r \in (0,1) \cap |K^\times|$ and consider,*

$$\Omega = \mathcal{R}_{\operatorname{Sp} \mathbb{T}_1}(f_0|f_1, \cdots, f_m) \in \mathfrak{Rat}_{\operatorname{Sp} \mathbb{T}_1}.$$

1. *If for all $i = 1, \cdots, m$, $N_r(f_i) \leq N_r(f_0)$, then there is $R \in (r, 1]$ such that $D(0,R) \subseteq \Omega \cup D(0,r)$.*

2. *If there is $i = 1, \cdots, m$ with $N_r(f_i) > N_r(f_0)$, then there is $R \in (r,1]$, such that $\Omega \cap D(0,R) \subseteq D(0,r)$.*

Furthermore, show that there is a van der Put point ξ such that $\xi \cap \mathfrak{Rat}_{\operatorname{Sp} \mathbb{T}_1}$ only consisting of rational opens satisfying the first condition.

Solution. Let r_i, R_i for $i = 0, \cdots, m$ be the corresponding constants in Problem 9.8. Pick the smallest R_i as R. If $\nu_r(f_i) = \nu_r(f_0)$ for any i, then as $\frac{|x|}{r} \geq 1$ and $j_r(f_i) \leq j_r(f_0)$, the inequality holds. If $\nu_r(f_i) < \nu_r(f_0)$, then we can take a small enough ρ such that

$$\forall x \in K^\circ, r < |x| < \rho, \nu_r(f_i) \left|\frac{x}{f_i}\right|^{j_r(f_i)} \leq \nu_r(f_0) \left|\frac{x}{f_0}\right|^{j_r(f_0)}.$$

Therefore, we may take smaller one from R, ρ, we obtain the first part.

Suppose there is $N_r(f_i) > N_r(f_0)$. Then there is $R \in (r,1]$ such that

$$\forall x \in K^\circ, r < |x| < R \Rightarrow |f_i(x)| > |f_0(x)|.$$

Therefore,

$$\Omega \cap (D(0,R) \backslash D(0,r)) = \emptyset.$$

Therefore, we obtain the statement. For ξ being a van der Put point. Clearly $X = \mathcal{R}(1|0) \in \xi$. Upward closure is clear from the condition. By the dichotomy of two conditions, taking smaller R, we obtain the closure under finite intersection. For the finite covering condition, is also clear from the dichotomy. □

Problem 10.2. *Let $\xi \in X^*$ and suppose $r := R_\xi \in (0,1) \cap |K^\times|$ satisfying the following. For all $R \in (r,1]$,*

1. *$0 \in D(\xi, R)$,*

2. *$D(\xi, r) \notin \xi$.*

Then ξ coincides with the one in Problem 10.1.

Solution. Suppose $0 \in D(\xi, R)$, then for any $y \in D(\xi, R)$,

$$|y| = |x - 0 + y - x| \leq R.$$

Therefore, $D(\xi, R) = D(0, R)$.

Note that it is enough to show that at the level of $\mathfrak{Rat}_X$. Let,

$$\Omega = \mathcal{R}_X(f_0|f_1, \cdots, f_m) \in \mathfrak{Rat}_X.$$

If the first condition in Problem 10.1 holds then there is $R \in (r, 1]$ such that

$$D(0, R) \subseteq \Omega \cup D(0, r).$$

By Problem 7.2 and that $r \in (0, 1) \cap |K^\times|$, we see $D(0, r) \in \mathfrak{Rat}_X$. Therefore by the finite covering condition, $\Omega \in \xi$.

Suppose $\Omega \cap D(0, R) \subseteq D(0, r)$. Then if $\Omega \in \mathfrak{Rat}_X$, then $D(0, r) \in \xi$ which is a contradiction. □

Definition 10.1. *Let $D \in \mathfrak{B}_{\leq R}$ and $x_0 \in K$. Then set,*

$$S(x_0, R) := \{y \in K \mid |x_0 - y| = R\},$$
$$\mathfrak{S}(D) = \{S(s, R) \mid x \in D\}.$$

Proposition 10.1. *There is a bijection*

$$\{B(y, R) \mid y \in D\} \overset{B(y,R) \mapsto D \backslash B(y,R)}{\longleftrightarrow} \mathfrak{S}(D).$$

Proof. Set $D = D(x, R)$. The map is well-defined as,

$$D \backslash B(y, R) = \{z \in D \mid |z - y| < R\}.$$

Then we have for $z \in D \backslash B(y, R)$,

$$R = |x - z| = |x - y + y - z| \leq \max\{|x - y|, |y - z|\}.$$

Therefore $|x - y| = R$. □

Problem 10.3. *Let $D \in \mathfrak{B}_{\leq R}$ and $\xi \in X^*$ be such that $D \in \xi$. Show that there exists at most one $S \in \mathfrak{S}(D)$ such that $S \notin \xi$.*

Solution. Let $B_1 = B(y_1, R), B_2 = B(y_2, R)$ be distinct open balls such that corresponding spheres S_1, S_2, respectively, fail to belongto ξ. If there is $z \in B_1 \cap B_2$, then

$$|y_1 - y_2| = |y_1 - z + z - y_2| < R.$$

Therefore, $B_1 = B_2$. Thus $B_1 \cap B_2 = \emptyset$. Since $\xi \subseteq \mathscr{O}_X$, and that $D = B_i \cup S_i \in \xi$, $S_i \notin xi$, implies $B_i \in \xi$. But the $\emptyset \in \xi$ which is a contradiction. □

Problem 10.4. *Suppose K is algebraically closed. Let $g \in \mathbb{T}_1$, show that the set,*

$$B(g) := K^\circ \backslash \{x \in K^\circ \mid |g(x)| < \|g|\mathbb{T}_1\|\}$$

is an intersection of finitely many elements from $\mathfrak{S}(K^\circ)$.

Solution. If $g = 0$, then $B(g) = K^\circ$ is the empty intersection. So we assume $g \neq 0$. Set,

$$g = \sum g_i T^i$$

and pick i_0 be the maximal index such that

$$|g_{i_0}| = \|g|\mathbb{T}_1\|.$$

Take $h = |g_{i_0}|^{-1} g$, then

$$|g(x)| < \|g|\mathbb{T}_1\| \iff |h(x)| < 1.$$

We then consider $k = K^\circ/K^{\circ\circ}$. Then as finitely many of coefficients h_i has absolute value 1 and others less than 1, we have

$$\overline{h} \in k[T].$$

Since K is algebraically closed, then so is k, thus,

$$h(T) = \prod (T - a_i).$$

Observe that

$$B(h) = \{x \in K^\circ \mid \overline{h(x)} = 0\}.$$

That is for any $x \in B(h)$, there is a_i such that $|x - a_i| < 1$. Therefore, we obtain,

$$B(h) = \bigcup B(a_i, 1).$$

Therefore,

$$K^\circ \backslash B(h) = \bigcap (D(0,1) \backslash B(a_i, 1)) = \bigcap S(a_i, 1).$$

□

Problem 10.5.

Sheet11.